ELEC5514 Lab5: LoRaWAN and Gateway

Starting this week, we will enter the world of LoRa communication. In this trial, we will deploy a LoRaWAN in the lab using a LoRa gateway. Students will use the LoRa transmitter to connect to the gateway and implement simple messaging.

Outline:

1. Introduce LoRa and LoRaWAN.
2. Modify the Endnote script and try to connect to the gateway.
3. (Optional) Realize the communication between the Lora board and the gateway.

Section 3 is an **optional** part of the LoRa communication experiment for next week.

Section 1

Introduction of LoRa Technology

LoRa (short for long range) is a spread spectrum modulation technique derived from chirp spread spectrum (CSS) technology. Semtech's LoRa devices and wireless radio frequency technology (LoRa Technology) is a long range, low power wireless platform that has become the de facto technology for Internet of Things (IoT) networks worldwide. It is a type of wireless telecommunication network designed to allow long range communications at a very low bit-rate and enabling long-life battery-operated sensors. LoRaWAN defines the communication and security protocol that ensures the interoperability with the Lora network.

For different type of applications, LoRa alliance specification provides three different service classes:

Class Name	**Intended Usage**
A-All	● Battery powered sensors or actuators with no latency constraint. ● Most energy efficient. ● Must be supported by all devices.
B-Beacon	● Battery powered actuators. ● Energy efficient communication class for latency-controlled downlink. ● Based on slotted communication synchronized with a network beacon.
C-Continuous	● Main powered actuators. ● Devices which can afford to listen continuously. ● No latency for downlink communication.

(The LoRa board used in our lab only support **Class A**.)

After the device is powered on, the code uses the finite state machine (FSM) to schedule the device to execute certain service, like joining a network, sending data or being low power mode (save the power). FSM uses different sate to represent the working mode and the state transition is triggered either by a timer event or by a radio event.

(The detail related to the LoRa protocol, please follow the document "LoRa standard overview" in folder Help Manual. You can also find other helpful documents in this folder. Please read them.)

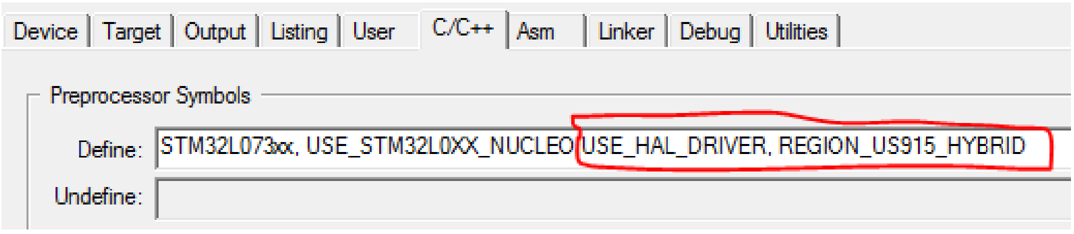
Section 2

Connect to Gateway

1. Connect board to PC.
2. Open the projects:

   SourceCode/Lab5_Lora/Projects/Multi/Applications/LoRa/End_Node/MDK-ARM/STM32l073RZ-Nucleo/lora

3. Configure the board. Add the following parameters in the Define/C/C++.



4. Open the file *Comissioning.h* and configure the following parameters:
   a. Set the OTTA activation procedure.
   b. Set your own LoRaWAN device EUI.

Group Name	**Device EUI**
Group 1	0x01,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 2	0x02,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 3	0x03,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 4	0x04,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 5	0x05,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 6	0x06,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 7	0x07,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 8	0x08,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 9	0x09,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 10	0x10,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 11	0x11,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 12	0x12,0x01,0x01,0x01,0x01,0x01,0x01,0x01
Group 13	0x13,0x01,0x01,0x01,0x01,0x01,0x01,0x01

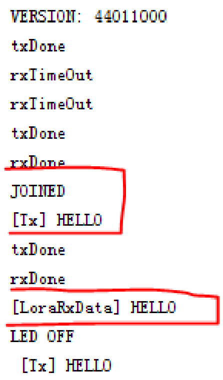
   c. Find the meaning of the LoRaWAN application EUI and LoRaWAN application key.
   d. Now you can try to connect your board to the gateway, and you can find the feedback that the connection is built in the serial monitor.

When you finish Section 2, call your tutor to check you result. You need to take the screenshot of your result for use in your report.

Section 3 (Optional)

Section 3 is an **optional** part of the LoRa communication experiment for next week.
In this section, you are asked to implement a simple communication between the Lora board and the gateway. You need to send your group number as a message to the gateway, like '*Group1*'. In this time, you can create this message directly in your code.

The following figures is an example by sending '*Hello*', and you also can use another way to achieve this function:



Tips: You can check the functions of *LoraTxData* and *LoraRxData* to understand the transmission and receive process. Note that the key word *"AppData"* has already been defined in this file and it is the set of the transmission information. You can read the codes from line around 219 for help, such as "*AppData->Port*", "*AppData->Buff*" and "*AppData->BuffSize*".

When you finish Section 3, call your tutor to check you result. You need to take the screenshot of your result for use in your report.